# Experiment 8: Fireworks and flame tests

Fireworks

**Notes**

### Skyshows explode around the country to celebrate Australia day.

The characteristic colours of **fireworks** are due to particular metal elements. In this case, the energy to excite the atoms comes from the exothermic chemical reaction between air and gunpowder. Almost immediately, excited atoms emit photons with characteristic wavelengths as the electrons fall from higher energy levels to lower energy levels. Copper gives a green display, sodium gives yellow, calcium gives red, and potassium gives violet, for example. This property of elements is also used in advertising, for example, in 'neon' signs. In industry we use this property to identify elements, and, using a 'flame test' in an atomic absorption spectroscope, AAS, we can even measure concentrations of specific elements.

Fireworks can be red, blue, green; all colours for that matter. Have you ever wondered why or how? You may have noticed the different flame colours that you see when a glossy magazine is burnt or that cooking salt spilt near a gas flame causes the flame to go an orangey yellow colour. The purpose of this experiment is to examine the colour produced by different salts when they are flame tested.

## Equipment

watch glass
matches
ethanol (10-20 mL)
tooth picks

small samples (pea sized) of salts such as:

copper(II) sulfate [$CuSO_4$]
potassium nitrate [$KNO_3$]
sodium chloride [$NaC\ell$]
barium nitrate [$Ba(NO_3)_2$]
iron(II) sulfate [$FeSO_4$]
strontium nitrate [$Sr(NO_3)_2$]
magnesium nitrate [$Mg(NO_3)_2$]
lithium nitrate [$LiNO_3$]
calcium nitrate [$Ca(NO_3)_2$]

## Procedure

1. Place a small amount of the salt in a clean watch glass.
2. Add 1-2 mL of ethanol to the sample and stir with a tooth pick.
3. Carefully light the ethanol.
4. When the flame burns low observe the colour produced.
5. Repeat for each salt using a clean watch glass each time.

## Processing of results and questions

1. What component of the salts is responsible for the colours observed?
2. What assumption is made about the ethanol in the flame test?
3. Why must a clean watch glass be used for each test?
4. Explain how the coloured light observed is produced.
5. Give a practical use of this type of test.
6. Often different colours are used to describe the same flame colour, e.g. lilac, mauve or purple for potassium.
   (a) Why might this be a problem?
   (b) Is this a random or systematic error? Justify your choice.

**SAFETY NOTE:**
- **Ethanol is flammable and it is essential that amounts used are as indicated. Keep the stock solution away from any naked flame.**
- **Do this on a heatproof mat.**